Result 1 (forward edge). [Topology foundations, forward orientation convention]
Let n be a natural number. Let d be a function from $\{0, \ldots, n-1\}$ to $\{\text{false}, \text{true}\}$. Let e be a vertex in $\{0, \ldots, n-1\}$. Assume $d(e) = \text{true}$ (`forward`). Then

$$s(e) = e \text{ and } t(e) = e + 1.$$

Result 2 (backward edge). [Topology foundations, backward orientation convention]
Let n be a natural number. Let d be a function from $\{0, \ldots, n-1\}$ to $\{\text{false}, \text{true}\}$. Let e be a vertex in $\{0, \ldots, n-1\}$. Assume $d(e) = \text{false}$ (`backward`). Then

$$s(e) = e + 1 \text{ and } t(e) = e.$$

Result 3 (constant diagram arrow). [Topology foundations, constant diagram component]
Let K be a type. Let *field* be a field structure on K. Let V and E be types. Let Q be a directed shape with vertices in V and edges in E. Let M be a module over K. Let e be an element of E. Then

$$\text{const}(M).\text{arrow}(e) = \text{id}_M .$$

Result 4 (interval vector support). [Bendich, Edelsbrunner, Morozov, and Patel, section 2.2, labelled interval basis]
Let K be a type. Let *field* be a field structure on K. Let n be a natural number. Let d be a function from $\{0, \ldots, n-1\}$ to $\{\text{false}, \text{true}\}$. Let Z be a module diagram on Q_d. Let I be a type. Let L be a family of ordered interval labels indexed by I on vertices 0 through n. Let B be an interval basis of Z with labels L. Let v be a vertex in $\{0, \ldots, n+1-1\}$. Let i be an element of I. Then

$$B_{v,i} \neq 0 \text{ if and only if } i \text{ belongs to } L_v.$$

Adapted from the cited source.

Result 5 (interval edge action). [Bendich, Edelsbrunner, Morozov, and Patel, section 2.2, maps on labelled intervals]
Let K be a type. Let *field* be a field structure on K. Let n be a natural number. Let d be a function from $\{0, \ldots, n-1\}$ to $\{\text{false}, \text{true}\}$. Let Z be a module diagram on Q_d. Let I be a type. Let L be a family of ordered interval labels indexed by I on vertices 0 through n. Let B be an interval basis of Z with labels L. Let e be a vertex in $\{0, \ldots, n-1\}$. Let i be an element of I. Assume i belongs to $L_{s(e)}$ (`active`). Then

$$Z.\text{arrow}(e)(B_{s(e),i}) = B_{t(e),i}.$$

Adapted from the cited source.

Result 6 (constant zigzag has interval basis). [Topology foundations, constant zigzag construction]
Let K be a type. Let *field* be a field structure on K. Let n be a natural number. Let d be a function from $\{0, \ldots, n-1\}$ to $\{\mathrm{false}, \mathrm{true}\}$. Let M be a module over K. Let I be a type. Let b be a basis of M indexed by I. Then

$$(\text{there exists } L : \mathcal{I},\ (\text{there exists } B : \mathrm{IBasis}(\mathrm{const}_d(M), L),\ (\text{for every } v : \{0, \ldots, n+1-1\},\ (\text{for every } i : I,\ i \text{ belongs to } L_v)))) .$$

LLM-assisted specimens and proofs (gpt-6, 2026-11-06). See PROVENANCE.md. This is not a proof of pyramid basis extension.